

Simultaneous Confidence Intervals

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Motivating parameter combinations

$$\begin{aligned} & \beta^{p+1} \quad (\beta_1 - \beta_2, \beta_2 - \beta_3) \in \mathbb{R}^2 \\ & \sim \\ & (\underline{\beta}_1, \underline{\beta}_2, \beta_3) \in \mathbb{R}^3 \end{aligned}$$

$$\left\{ \begin{aligned} & \underline{a}_1 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \underline{a}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad r = 2 \\ \\ & \underline{a}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \underline{a}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \underline{a}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad r = 3 \end{aligned} \right.$$

General linear combinations:

$$\underline{a}_i^T \beta, \quad i = 1, \dots, r.$$

We want,

$$\{L_i(Y), U_i(Y)\}, \quad L_i(Y) \leq U_i(Y) \quad \forall i, \text{ w.p. } 1$$

Definition of Simultaneous Coverage Requirement

$$\begin{aligned} & \inf_{\beta} P_{\beta} \left(\underbrace{L_1(Y) \leq \underline{a}_1^T \beta \leq U_1(Y)}_{E_1}, \dots, \underbrace{L_r(Y) \leq \underline{a}_r^T \beta \leq U_r(Y)}_{E_r} \right) = (1 - \alpha) \\ & \Rightarrow P_{\beta} (E_1 \cap E_2 \cap \dots \cap E_r) \geq (1 - \alpha) \quad \forall \beta. \end{aligned}$$

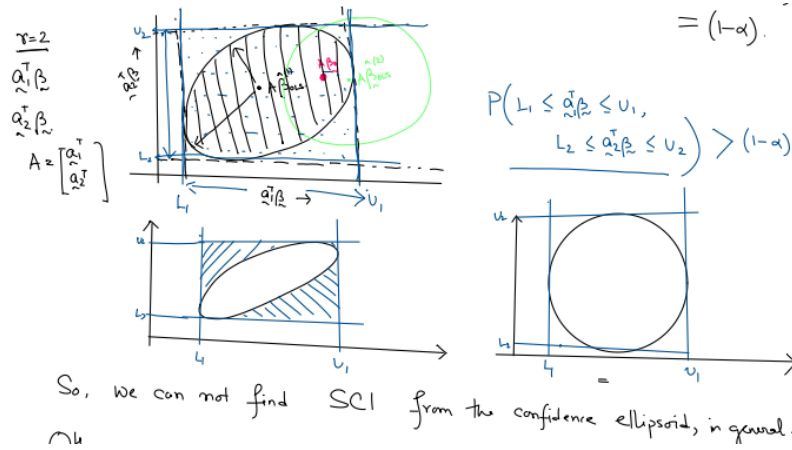
Confidence Ellipsoid

Recall, confidence ellipsoids:

$$P_{\tilde{\beta}} \left((A\hat{\tilde{\beta}}_{\text{OLS}} - A\tilde{\beta})^T \left[\underline{A(X'X)^{-1}A'} \right]^{-1} (A\hat{\tilde{\beta}}_{\text{OLS}} - A\tilde{\beta}) \leq r \frac{\text{RSS}}{n-p-1} F_{\alpha; r, n-p-1} \right) = (1-\alpha).$$

For illustration with $r = 2$:

$$r = 2 \quad \begin{matrix} a^T \beta \\ \sim_1 \end{matrix} \quad \begin{matrix} a^T \beta \\ \sim_2 \end{matrix} \quad A = \begin{bmatrix} a^T \\ \sim_1^T \\ a^T \\ \sim_2^T \end{bmatrix}$$



$$P \left(L_1 \leq \tilde{a}_1^T \tilde{\beta} \leq U_1, L_2 \leq \tilde{a}_2^T \tilde{\beta} \leq U_2 \right) > (1 - \alpha)$$

So, we can not find SCI from the confidence ellipsoid, in general.

Using

$$\begin{aligned} \begin{matrix} a^T \hat{\tilde{\beta}} \\ \sim \end{matrix} &\sim N \left(\begin{matrix} a^T \beta \\ \sim \end{matrix}, \sigma^2 \begin{matrix} a^T (X'X)^{-1} a \\ \sim \end{matrix} \right) \\ \underbrace{P_{\tilde{\beta}} \left(\begin{matrix} a^T \hat{\tilde{\beta}} \\ \sim \end{matrix} - \sqrt{\frac{\text{RSS}}{n-p-1}} \sqrt{\begin{matrix} a^T (X'X)^{-1} a \\ \sim \end{matrix}} \cdot t_{\alpha/2; n-p-1} < \right.}_{L} \\ \left. \begin{matrix} a^T \beta \\ \sim \end{matrix} \leq \begin{matrix} a^T \hat{\tilde{\beta}} \\ \sim \end{matrix} + \sqrt{\frac{\text{RSS}}{n-p-1}} \sqrt{\begin{matrix} a^T (X'X)^{-1} a \\ \sim \end{matrix}} \cdot t_{\alpha/2; n-p-1} \right) = (1 - \alpha)}_{U} \end{aligned}$$

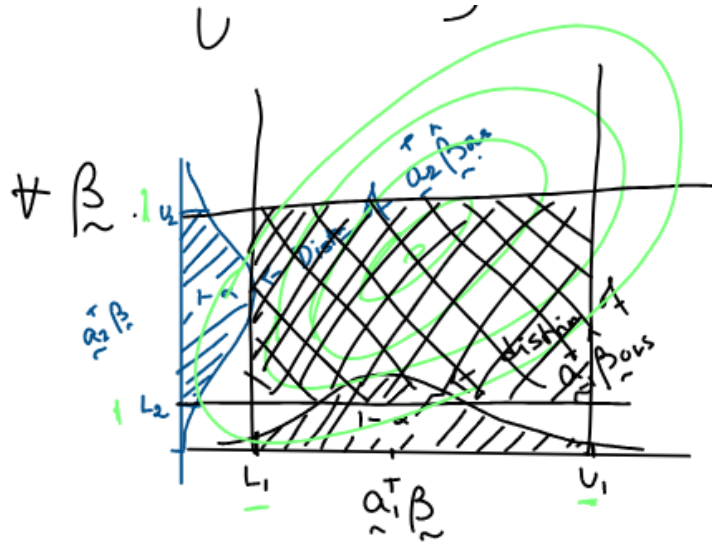
For each $j = 1, \dots, r$, we have

Individual CI Coverage

$$P_{\tilde{\beta}} \left(\underbrace{L_j(Y) < \begin{matrix} a^T \beta \\ \sim_j \end{matrix} \leq U_j(Y)}_{E_j} \right) = (1 - \alpha) \quad \forall \tilde{\beta}.$$

What is:

$$\begin{aligned}
 & P_{\beta}(E_1 \cap E_2 \cap \dots \cap E_r) \\
 &= P_{\beta} \left(\bigcap_{j=1}^r E_j \right) = 1 - P_{\beta} \left(\left(\bigcap_{j=1}^r E_j \right)^c \right) \\
 &= 1 - P_{\beta} \left(\bigcup_{j=1}^r E_j^c \right) \geq 1 - \sum_{j=1}^r P_{\beta}(E_j^c) = 1 - r\alpha.
 \end{aligned}$$



Example

Ex: $r = 10$; $\alpha = 0.05$.

$$P_{\beta}(E_1 \cap \dots \cap E_r) \geq 0.5$$

Also, if these events $E_1, \dots, E_r$ are independent,

$$P_{\beta}(E_1 \cap \dots \cap E_r) = \prod_{j=1}^r P_{\beta}(E_j) = (1 - \alpha)^r.$$

Example Independent Case

Ex: $r = 10$; $\alpha = 0.05$, $P_{\beta}(E_1 \cap \dots \cap E_r) \approx 0.6$.

We have to make sure that,

$$P_{\beta}(E_1 \cap \dots \cap E_r) \geq (1 - \alpha)$$

Bonferroni's correction

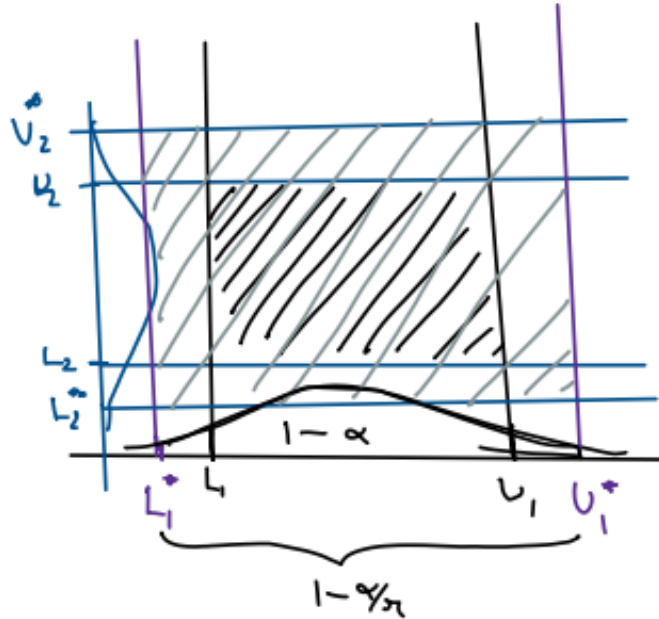
Bonferroni suggested taking the level of significance to be $\frac{\alpha}{r}$ instead of α , (confidence

coefficient) $(1 - \frac{\alpha}{r}) (1 - \alpha)$, for the individual confidence intervals.

$$P_{\tilde{\beta}}(E_j) = 1 - \frac{\alpha}{r} \quad \forall j = 1, \dots, r.$$

$$\Rightarrow P_{\tilde{\beta}}(E_1 \cap \dots \cap E_r) \geq 1 - r \cdot \frac{\alpha}{r} = (1 - \alpha) \quad \forall \tilde{\beta}.$$

If r is large then $\frac{\alpha}{r}$ is very small, so the resultant SCI is over-inclusive.



Scheffé's S-method

Suppose we want to have SCI for

$$a_{\tilde{1}}^T \beta, \dots, a_{\tilde{r}}^T \beta.$$

WLOG assume that $\{a_{\tilde{1}}, \dots, a_{\tilde{r}}\}$ are linearly independent.

$$H_0 : \begin{array}{c} a_{\tilde{1}}^T \beta = 0 \\ \vdots \\ a_{\tilde{r}}^T \beta = 0 \end{array}$$

$$A_{r \times p+1} = \begin{bmatrix} a'_{\tilde{1}} \\ \vdots \\ a'_{\tilde{r}} \end{bmatrix} \text{ has rank } r.$$

Recall

$$A_{\tilde{\text{OLS}}} \hat{\beta} \sim N_r \left(A_{\tilde{\text{OLS}}} \beta, \sigma^2 A_{\tilde{\text{OLS}}} (X'X)^{-1} A_{\tilde{\text{OLS}}} \right) \text{ and } \text{RSS} \sim \sigma^2 \chi_{n-p-1}^2 \text{ independently}$$

$$P_{\beta} \left(\underbrace{[A(\hat{\beta}_{\sim\text{OLS}} - \beta)]^T}_{\tilde{b}^T} \underbrace{[A(X'X)^{-1}A']^{-1}}_{L^{-1}} \underbrace{A(\hat{\beta}_{\sim\text{OLS}} - \beta)}_{\tilde{b}} \leq \underbrace{r \frac{\text{RSS}}{n-p-1} \cdot F_{\alpha; r, n-p-1}}_{\mathcal{E}_{\alpha}} \right) = (1 - \alpha) \quad \forall \beta_{\sim}$$

$$\Rightarrow P_{\beta} \left(\underbrace{\tilde{b}^T L^{-1} \tilde{b}}_{\sim} \leq \mathcal{E}_{\alpha} \right) = (1 - \alpha). \quad - (\star)$$

Key Claim

Claim:

$$\tilde{b}^T L^{-1} \tilde{b} = \sup_{\tilde{w} \neq 0} \frac{(\tilde{w}^T \tilde{b})^2}{\tilde{w}^T L \tilde{w}}$$

Proof of Claim

$$\begin{aligned} : \text{RHS} &= \sup_{\tilde{w} \neq 0} \frac{\tilde{w}^T \tilde{b} \tilde{b}^T \tilde{w}}{\tilde{w}^T L \tilde{w}} \\ &\left[\text{Define } \tilde{w}^* = L^{1/2} \tilde{w} \Leftrightarrow \tilde{w} = L^{-1/2} \tilde{w}^* \text{ where } L = \underbrace{A(X'X)^{-1}A'}_{\text{p.d. matrix}} \right] \\ &= \sup_{\tilde{w}^* \neq 0} \frac{\tilde{w}^{*T} L^{-1/2} \tilde{b} \tilde{b}^T L^{-1/2} \tilde{w}^*}{\tilde{w}^{*T} \tilde{w}^*} \\ &= \lambda_{\max} \left(\underbrace{L^{-1/2} \tilde{b} \tilde{b}^T L^{-1/2}}_{\sim} \right) = \tilde{b}^T L^{-1} \tilde{b}. \end{aligned}$$

From $(\star)$ we get:

$$P_{\beta} \left(\sup_{\tilde{w} \neq 0} \frac{(\tilde{w}^T \tilde{b})^2}{\tilde{w}^T L \tilde{w}} \leq \mathcal{E}_{\alpha} \right) = (1 - \alpha) \quad \forall \beta_{\sim}$$

$$\Rightarrow P_{\beta} \left(\underbrace{\tilde{w}^T A(\hat{\beta}_{\sim\text{OLS}} - \beta)}_{\sim} \leq \pm \sqrt{\tilde{w}^T L \tilde{w} \cdot \mathcal{E}_{\alpha}} \quad \forall \tilde{w} \neq 0 \right) \geq (1 - \alpha)$$

$$\left| \begin{array}{l} \tilde{b} = A\hat{\beta}_{\sim\text{OLS}} - A\beta. \\ L = A(X'X)^{-1}A' \end{array} \right|$$

In particular, taking w as $e_{\tilde{\sim}_1}, \dots, e_{\tilde{\sim}_r}$, we get:

$$P_{\beta_{\tilde{\sim}}} \left(\underbrace{a_{\tilde{\sim}_1 \sim \text{OLS}}^T \hat{\beta}_{\tilde{\sim}_1} - a_{\tilde{\sim}_1 \sim}^T \beta_{\tilde{\sim}_1}}_{w=e_{\tilde{\sim}_1}} \leq \pm \sqrt{\mathcal{E}_\alpha \cdot l_{11}}, \dots, \overbrace{a_{\tilde{\sim}_r \sim \text{OLS}}^T \hat{\beta}_{\tilde{\sim}_r} - a_{\tilde{\sim}_r \sim}^T \beta_{\tilde{\sim}_r}}^{w=e_{\tilde{\sim}_r}} \leq \pm \sqrt{l_{rr} \cdot \mathcal{E}_\alpha} \right) \geq (1 - \alpha)$$